Introduction to Spatial Modeling

from *ASMwSPDE*... by KGBLCSLR

Presentation by Peter Gao

Before we start...

Welcome to the Space-time reading group!

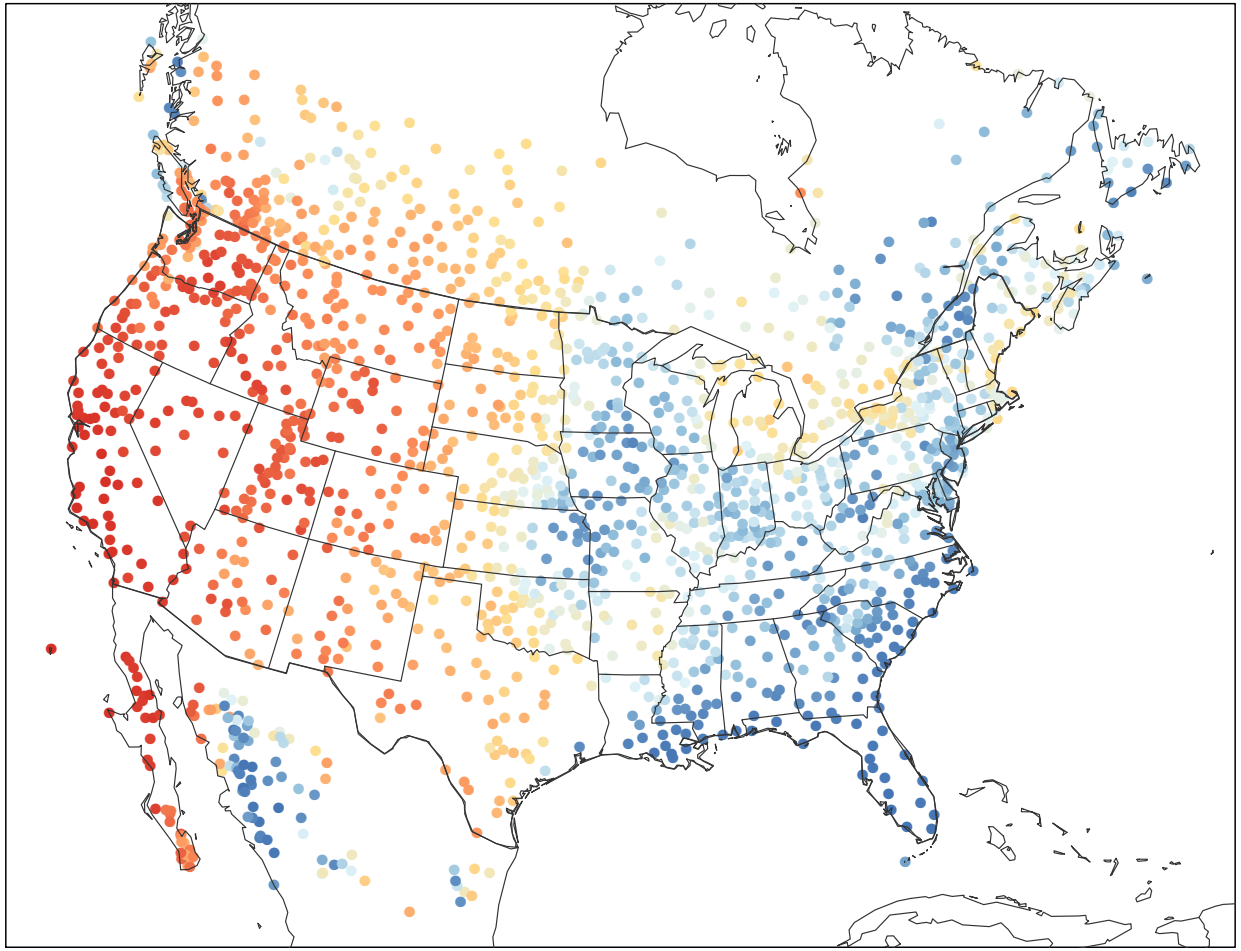
This quarter we'll be reading

Advanced Spatial Modeling with Stochastic Partial Differential Equations Using R and INLA by Elias T. Krainski, Virgilio Gómez-Rubio, Haakon Bakka, Amanda Lenzi, Daniela Castro-Camilo, Daniel Simpson, Finn Lindgren and Håvard Rue

We hope that everyone will be able to learn and participate regardless of background/familiarity with spatial statistics.

Please reach out with questions/comments/suggestions!

North American rainfall data



Douglas Nychka, Reinhard Furrer, John Paige and Stephan Sain (2017). fields: Tools for spatial data. doi: 10.5065/D6W957CT, R package version 9.6, www.image.ucar.edu/ nychka/Fields.

Modeling continuous spatial processes

Given spatially distributed observations, how do we model correlation and make predictions?

Let y be a continuous real-valued spatial process–for example, a function that returns a temperature value for any location in our spatial domain.

Modeling continuous spatial processes

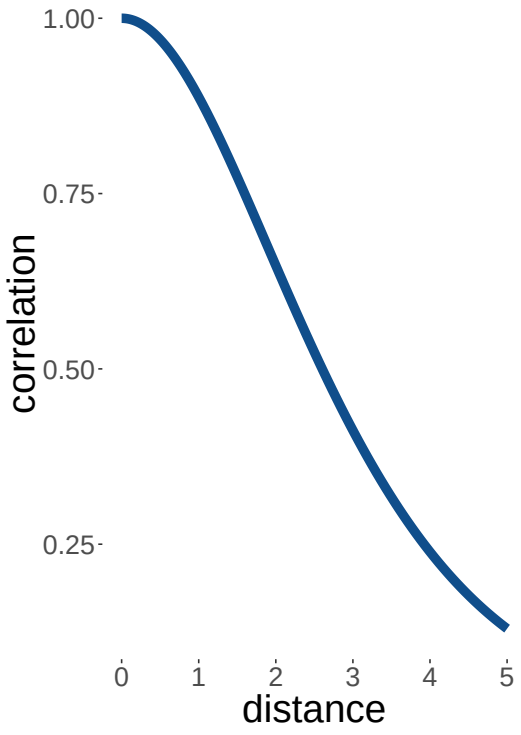
Gaussian process model:

A spatial process $y : \mathbb{R}^2 \to \mathbb{R}$ is a Gaussian process if for any finite set of locations $(x_1, \ldots, x_n)$, the observations $(y(x_1), \ldots, y(x_n))^T$ follow a multivariate normal distribution with mean function $\mu(x)$ and covariance function $\mathcal{C}(x, x')$.

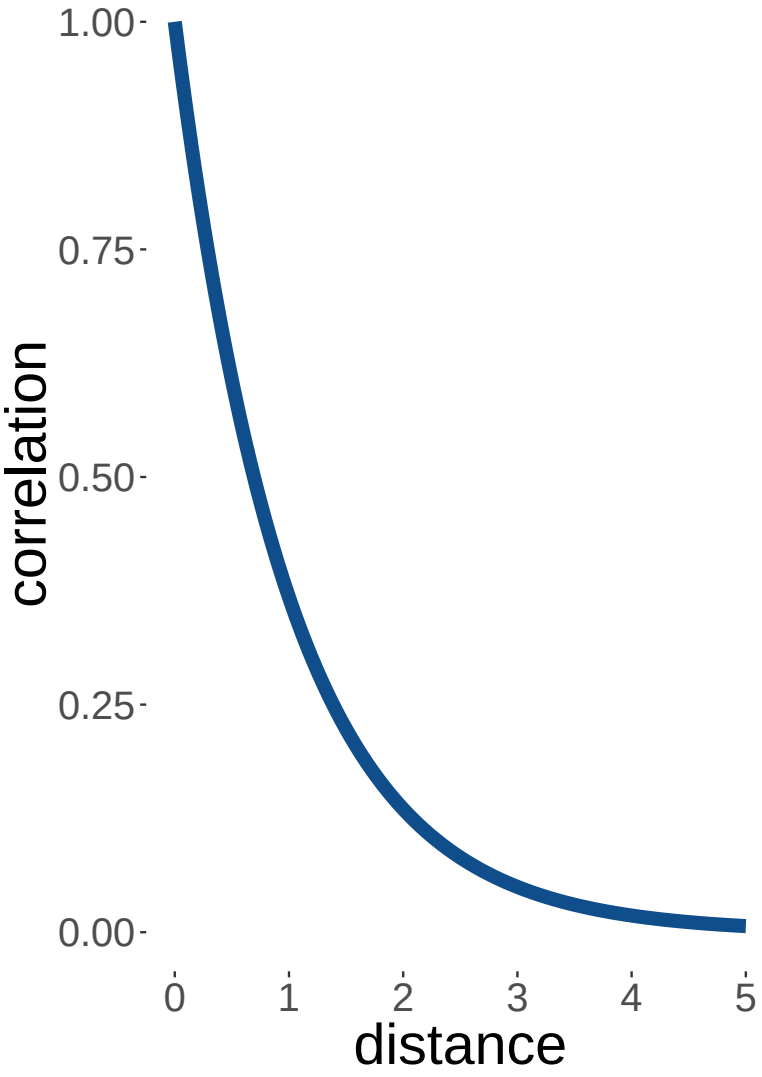
Specifying covariance function $\mathcal{C}$

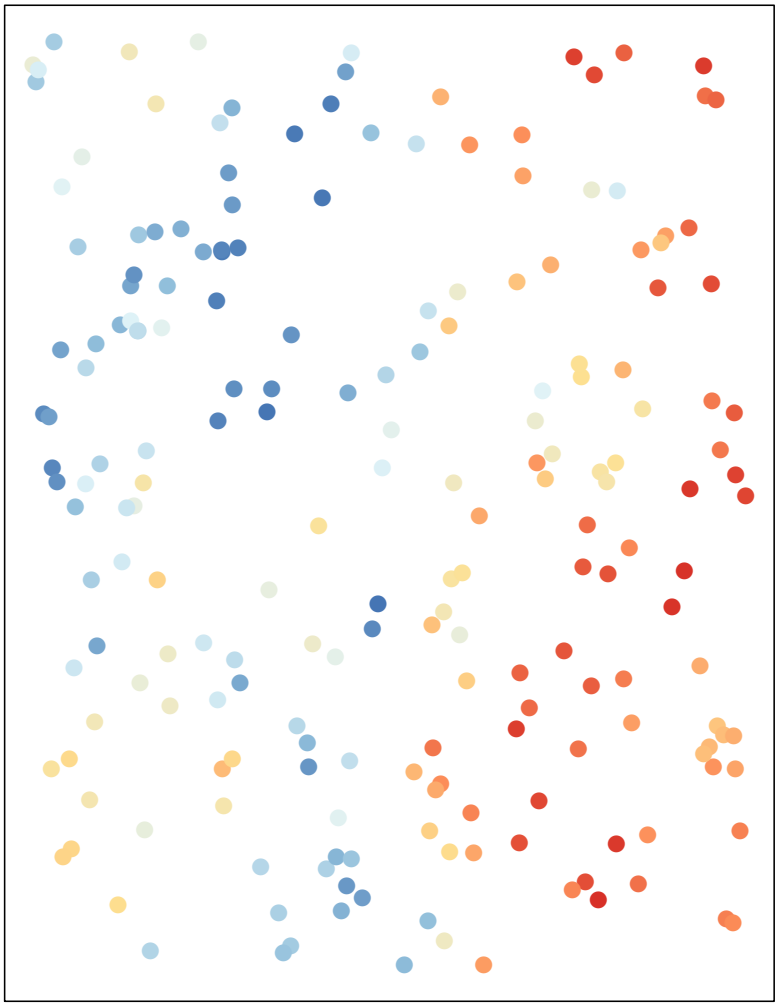
Assume that y:

- is **weakly stationary**: $\mathcal{C}(x, x')$ is a function of $x - x'$
- is **weakly isotropic**: $\mathcal{C}(x, x')$ is a function of $||x - x'||$

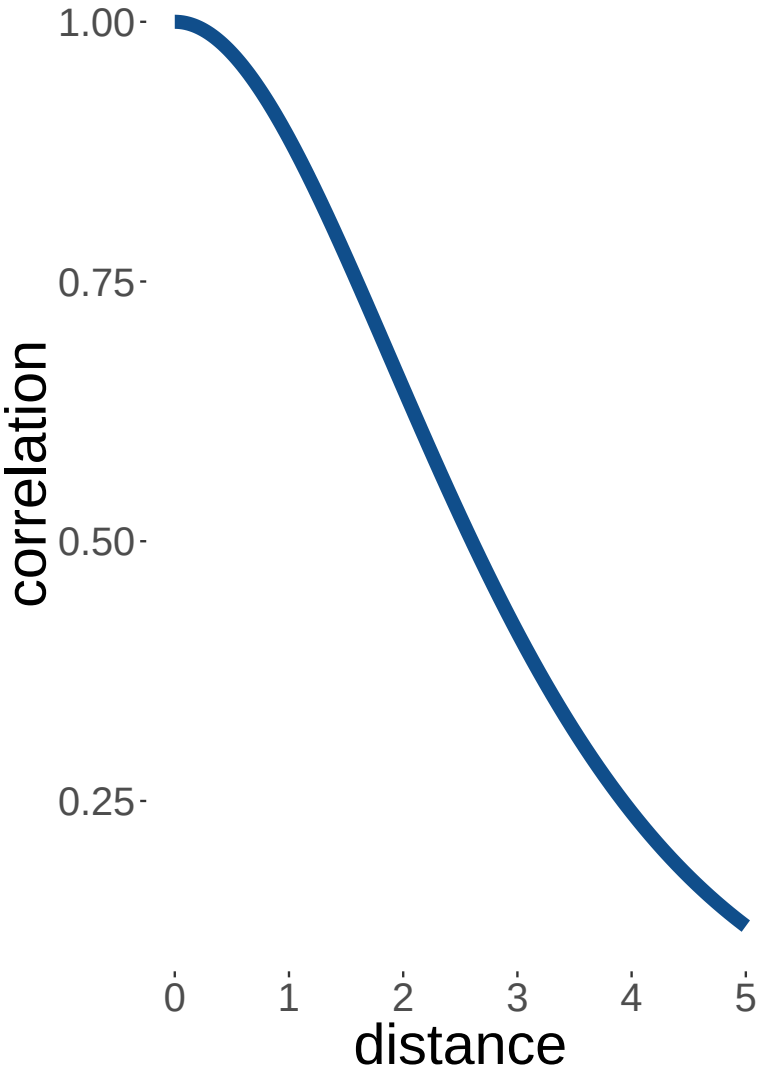


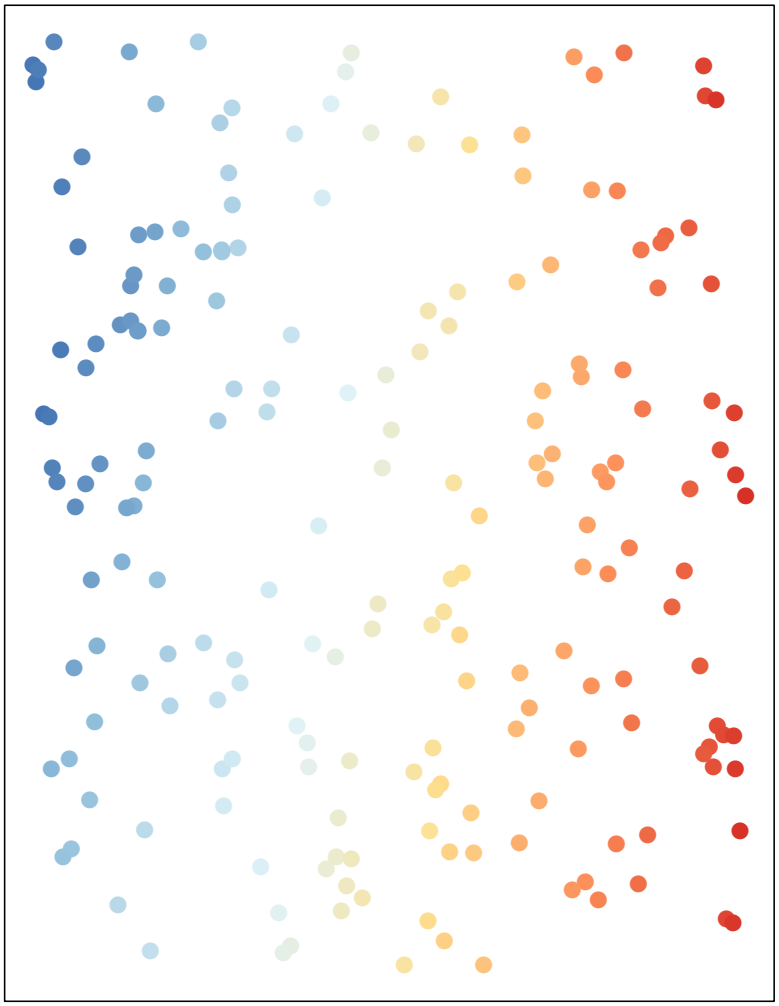
Gaussian process realization





Gaussian process realization





Matérn family

Matérn covariance:

$$\mathcal{C}(x, x') = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa||x - x'||)^\nu K_\nu(\kappa||x - x'||)$$

- σ^2 is a parameter controlling marginal variance
- ν is a smoothness parameter
- κ is a "range" parameter

1D Matérn processes

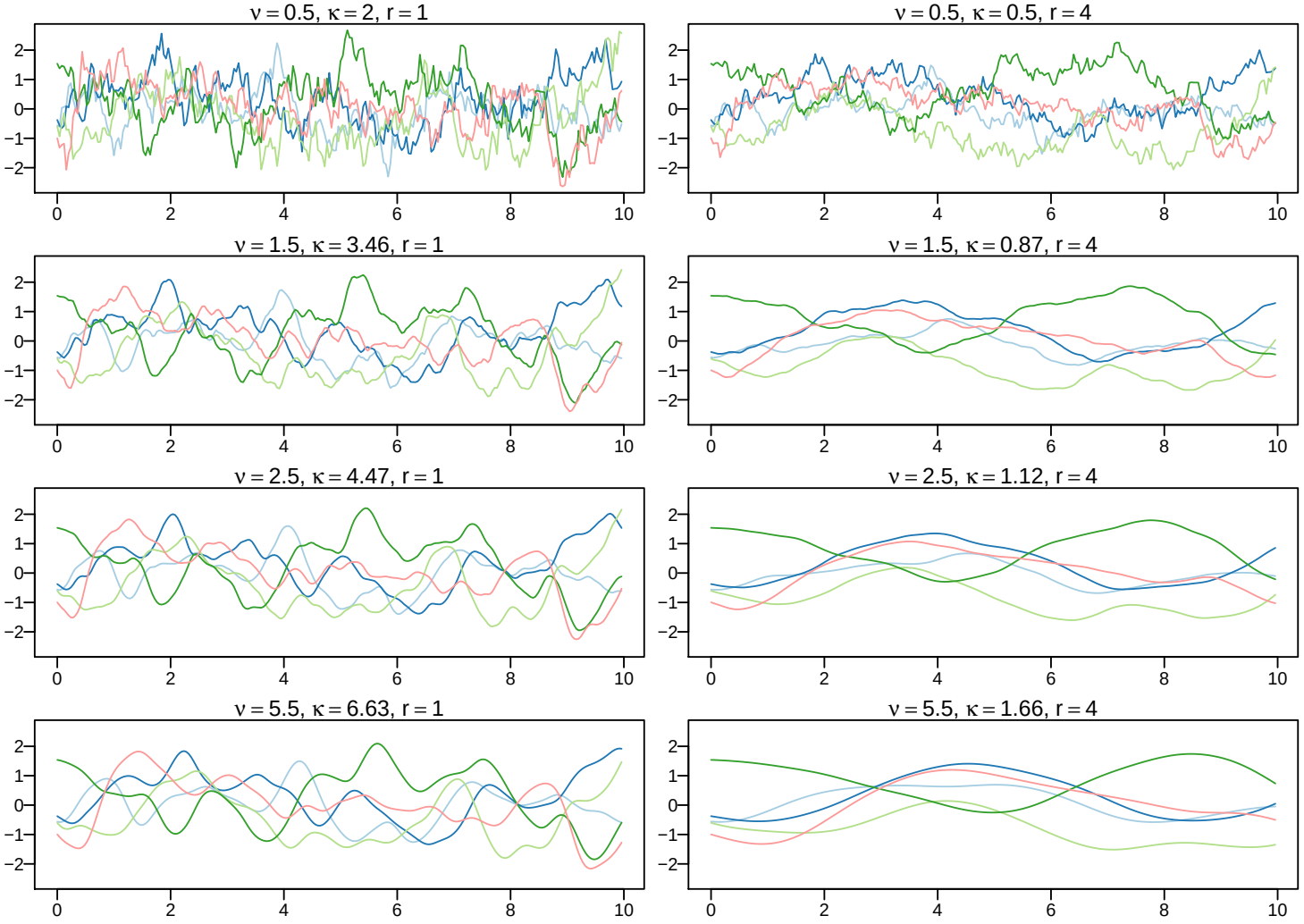


Figure 2.2 from *ASMwSPDE...*

Kriging for $y(\mathbf{x}_0)$

Suppose we want to predict the precipitation at a new location x_0, which we denote $y(x_0)$.

Assume that we have observations $(x_1, y_1), \ldots, (x_n, y_n)$ and denote $\mathbf{y} = (y_1, \ldots, y_n)^T$ and $\mathbf{x} = (x_1, \ldots, x_n)^T$

Kriging for $y(\mathbf{x}_0)$

Gaussian Process Model:

$$\mathbf{y} \sim N(\mathbf{Z}\boldsymbol{\beta}, \boldsymbol{\Sigma} + \sigma^2\mathbf{I})$$

- $\mathbf{Z}$: covariates matrix (e.g. elevation)
- $\boldsymbol{\beta}$: vector of coefficients (fixed effects)
- $\boldsymbol{\Sigma} = \left[\mathcal{C}(x_i, x_j)\right]_{i,j}$: spatial component of variance
- σ^2: variance from measurement errors

Kriging for $y(\mathbf{x}_0)$

Gaussian Process Model:

$$\mathbf{y} \sim N(\mathbf{Z}\boldsymbol{\beta}, \boldsymbol{\Sigma} + \sigma^2\mathbf{I})$$

Suppose we want to predict an observation $y(\mathbf{x}_0)$ at a new location $\mathbf{x}_0$.

Assume our covariance function $\mathcal{C}$ is correctly specified (so $\boldsymbol{\Sigma}$ is known) and that $\boldsymbol{\beta}$ is known.

Kriging for $y(\mathbf{x}_0)$

Then we have

$$\begin{bmatrix} \mathbf{y} \\ y(x_0) \end{bmatrix} \sim N\left(\begin{bmatrix} \mathbf{Z}\boldsymbol{\beta} \\ Z(x_0)\boldsymbol{\beta} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma} + \sigma^2\mathbf{I} & \boldsymbol{\eta}_0 \\ \boldsymbol{\eta}_0 & \mathcal{C}(x_0, x_0) + \sigma^2\mathbf{I} \end{bmatrix} \right)$$

Notation:

- $Z(x_0)$: covariates for new location x_0
- $\boldsymbol{\eta}_0 := (\mathcal{C}(x_1, x_0), \ldots, \mathcal{C}(x_n, x_0))^T$

The **best linear unbiased predictor** for y' (the kriging estimate) is the conditional mean:

$$\hat{y}(x_0) = E(y' \mid \mathbf{y}) = \mathbf{Z}(x_0)\boldsymbol{\beta} + \boldsymbol{\eta}_0(\boldsymbol{\Sigma} + \sigma^2\mathbf{I})^{-1}(\mathbf{y} - \mathbf{Z}\boldsymbol{\beta})$$

Two (related) challenges

Model:

$$\mathbf{y} \sim N(\mathbf{Z}\boldsymbol{\beta}, \boldsymbol{\Sigma} + \sigma^2\mathbf{I})$$

1. *Modeling covariance structure*

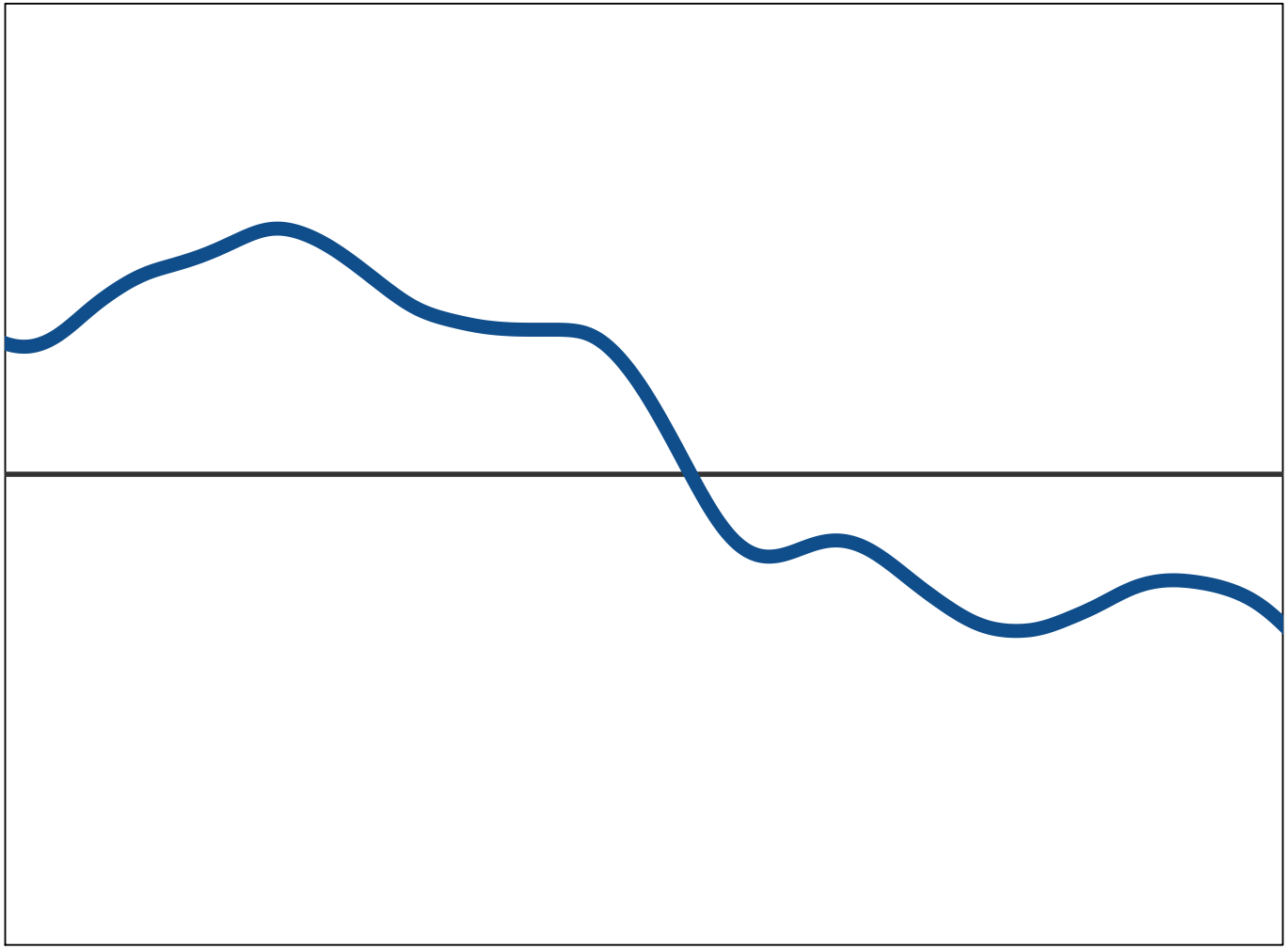
   We need to estimate $\mathcal{C}$ (which we use to compute $\boldsymbol{\Sigma}$).

2. *Ensuring computational efficiency*

   Evaluating the likelihood and making predictions requires computation of determinant and Cholesky decomposition of $n \times n$ matrix $\boldsymbol{\Sigma} + \sigma^2\mathbf{I}$ (in general, $\mathcal{O}(n^3)$ operations)

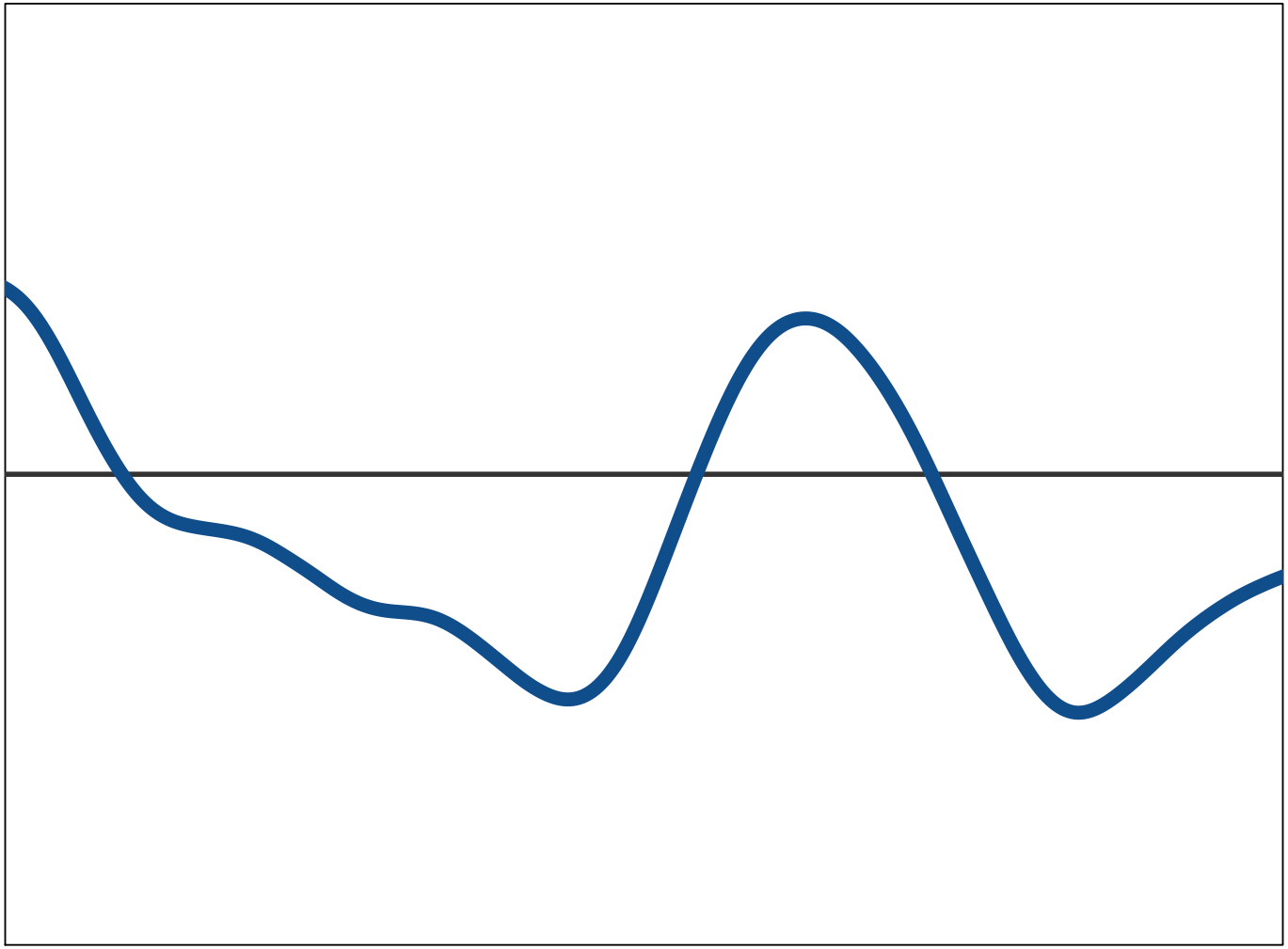
Approximating a 1-D Gaussian process

One sample from a Gaussian process with unknown covariance:



Approximating a 1-D Gaussian process

and another sample from the same process:

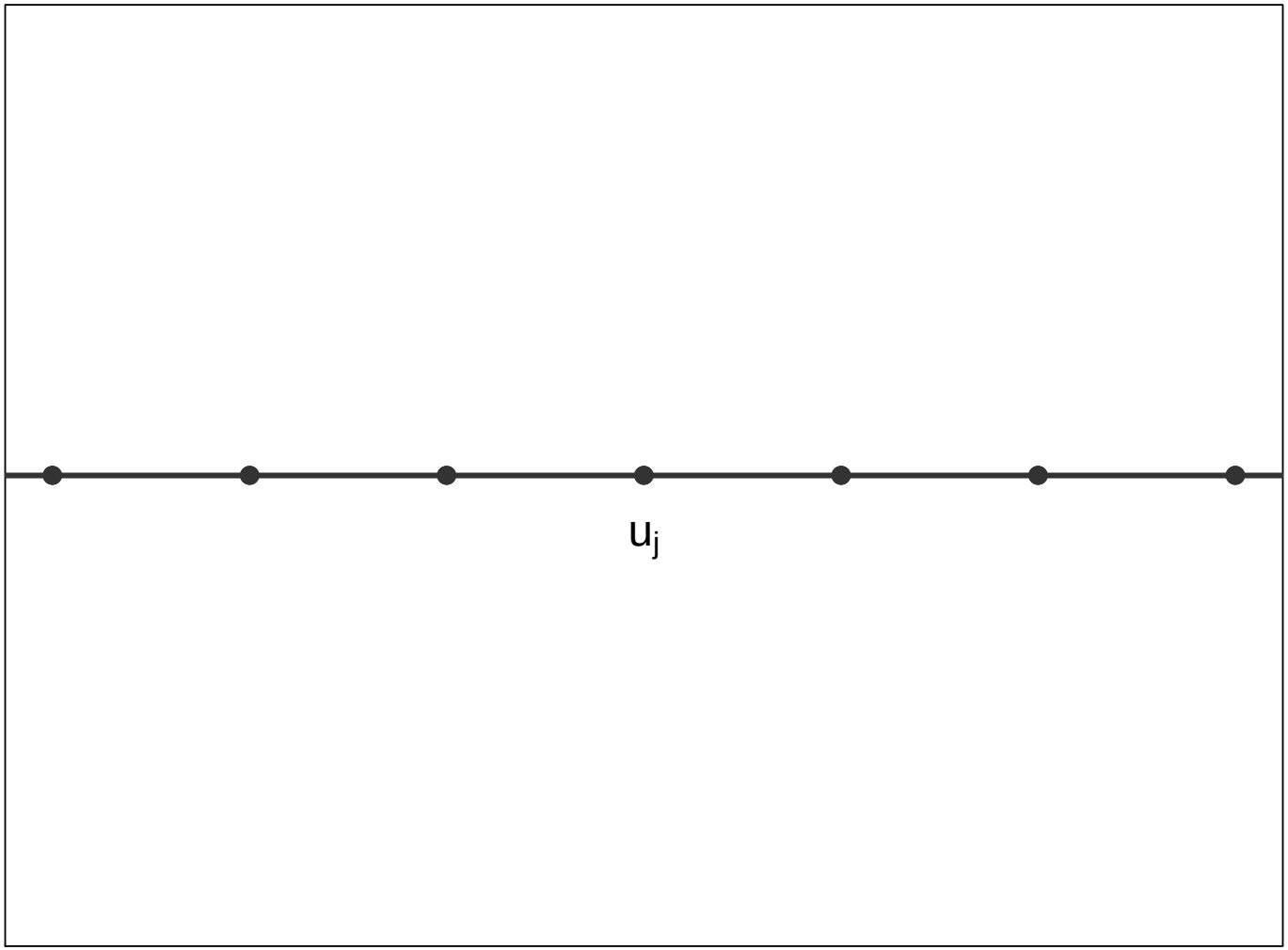


Approximating a 1-D Gaussian process

We want to construct a Gaussian process $y(x)$ that produces samples that "look like" our observations.

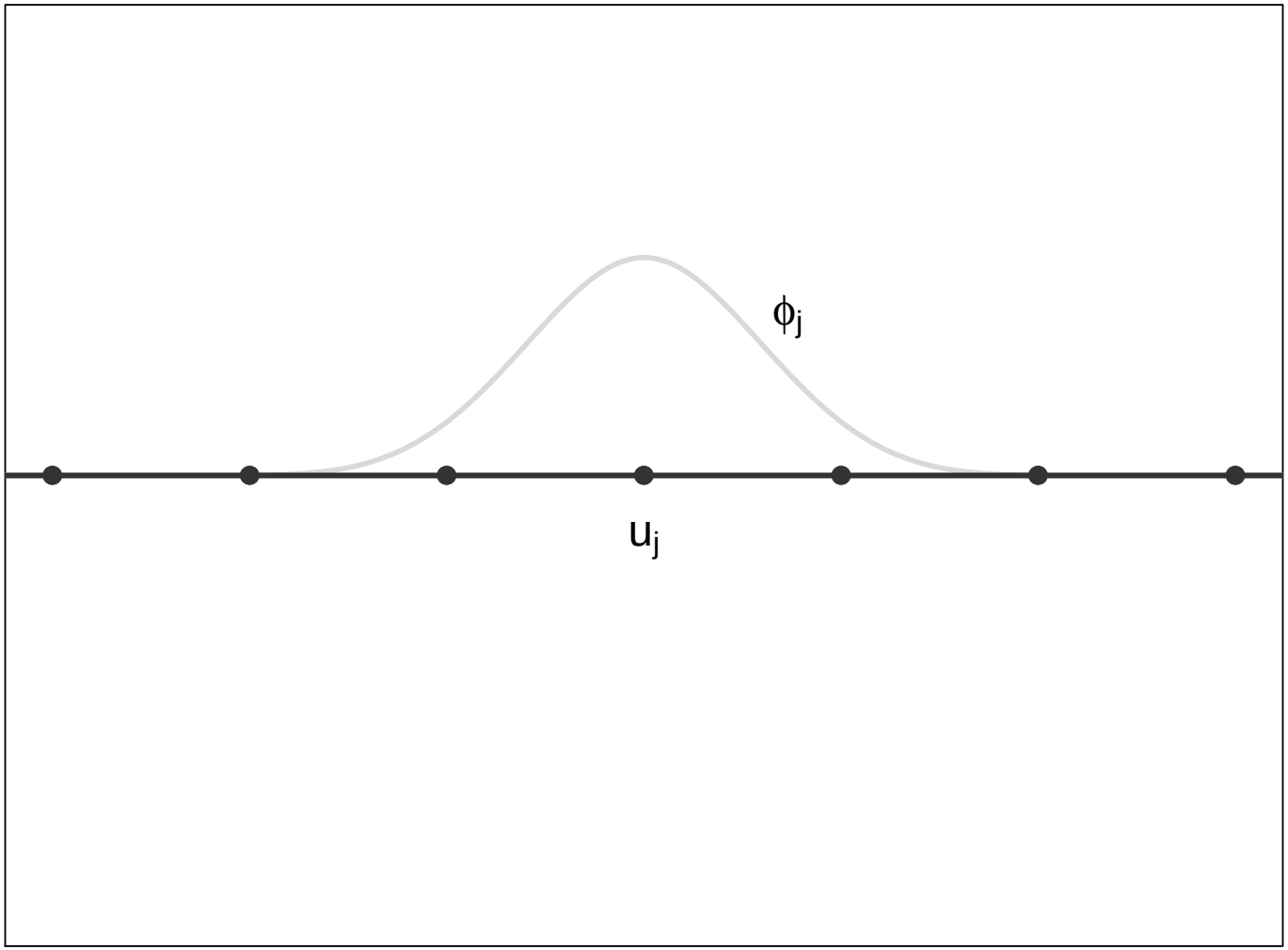
Approximating a 1-D Gaussian process

Draw a grid of points $\{u_j\}$ on our domain:



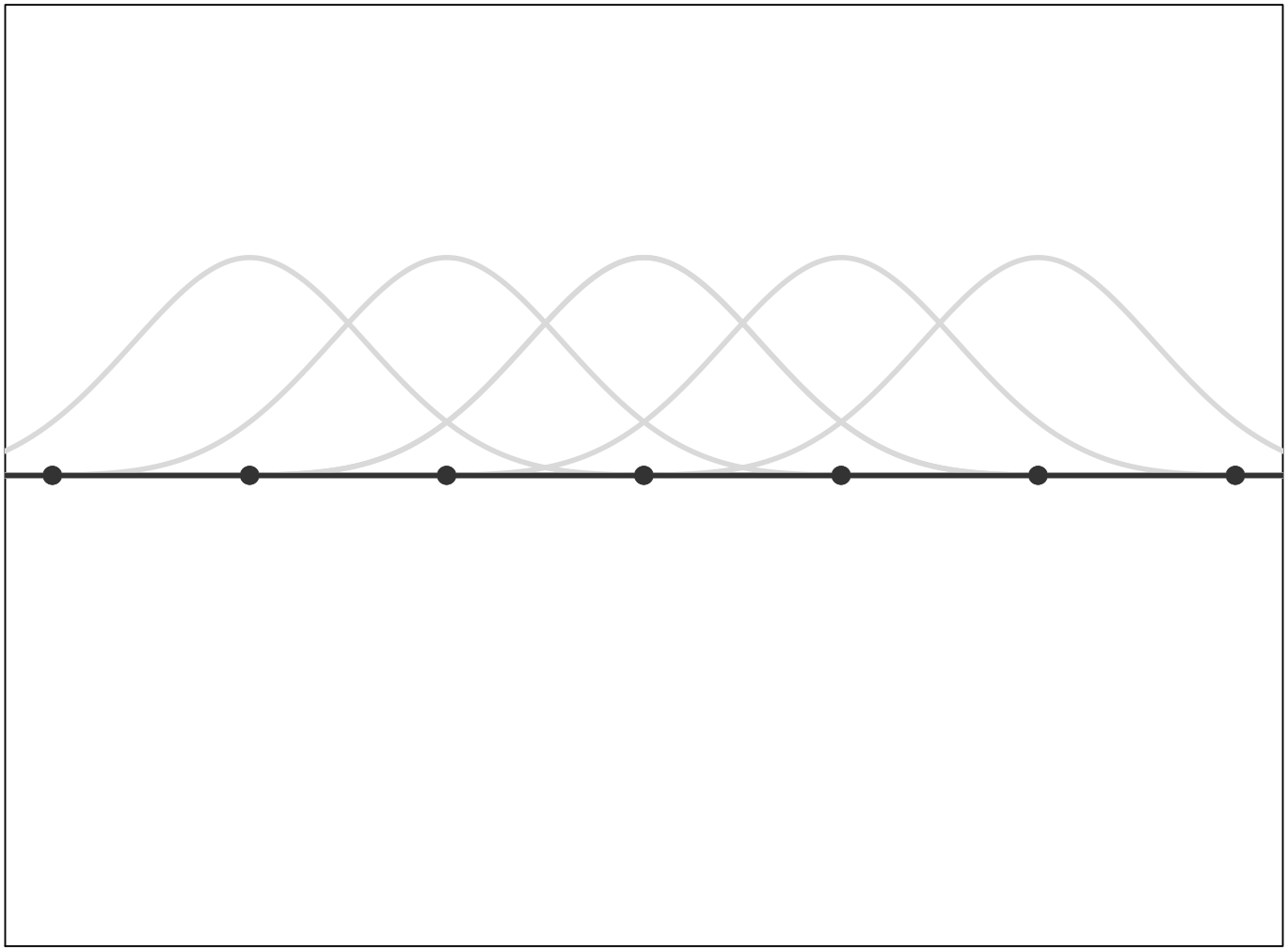
Approximating a 1-D Gaussian process

At each u_j, place a basis function ψ_j:



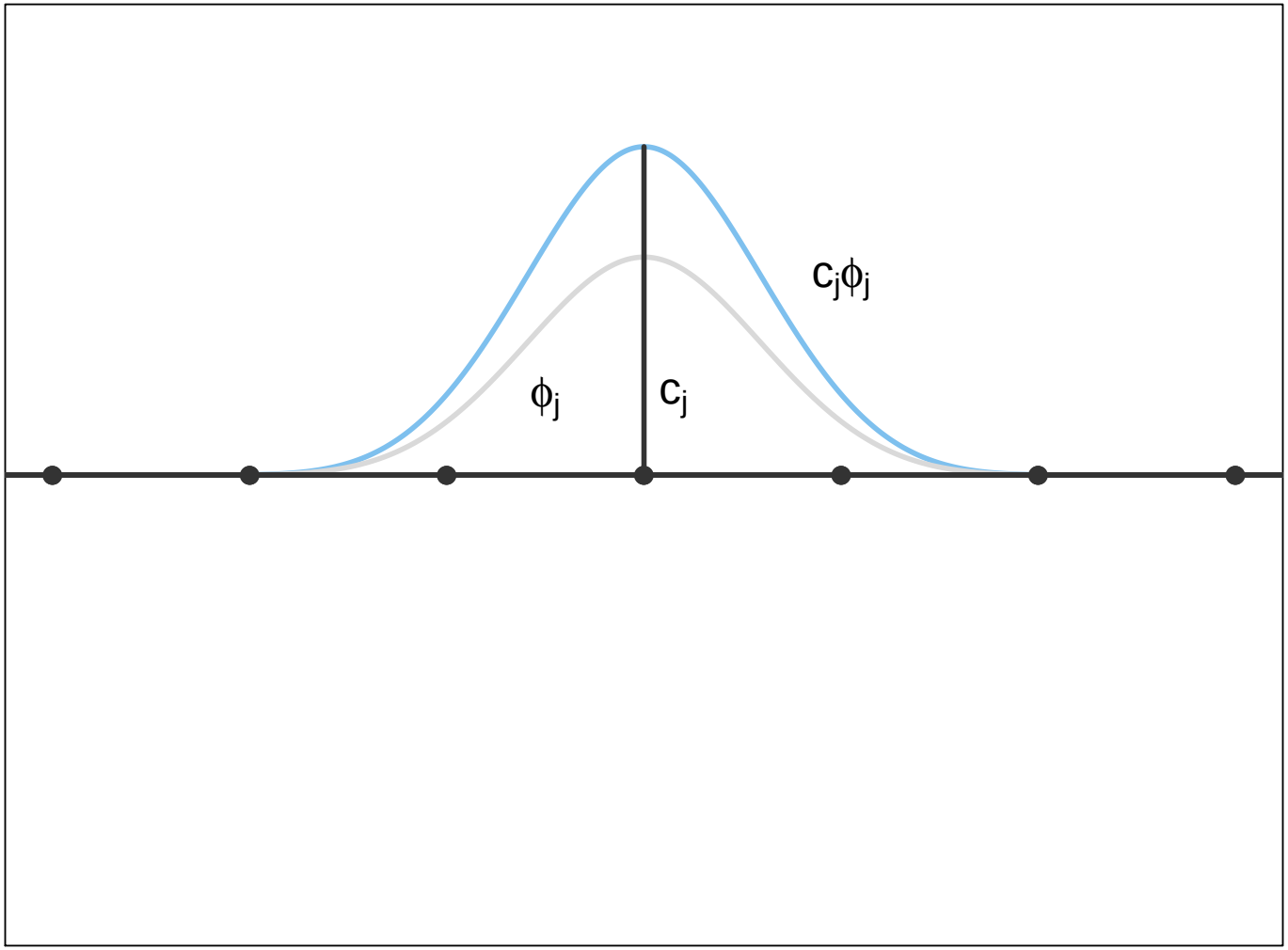
Approximating a 1-D Gaussian process

At each u_j, place a basis function ψ_j:



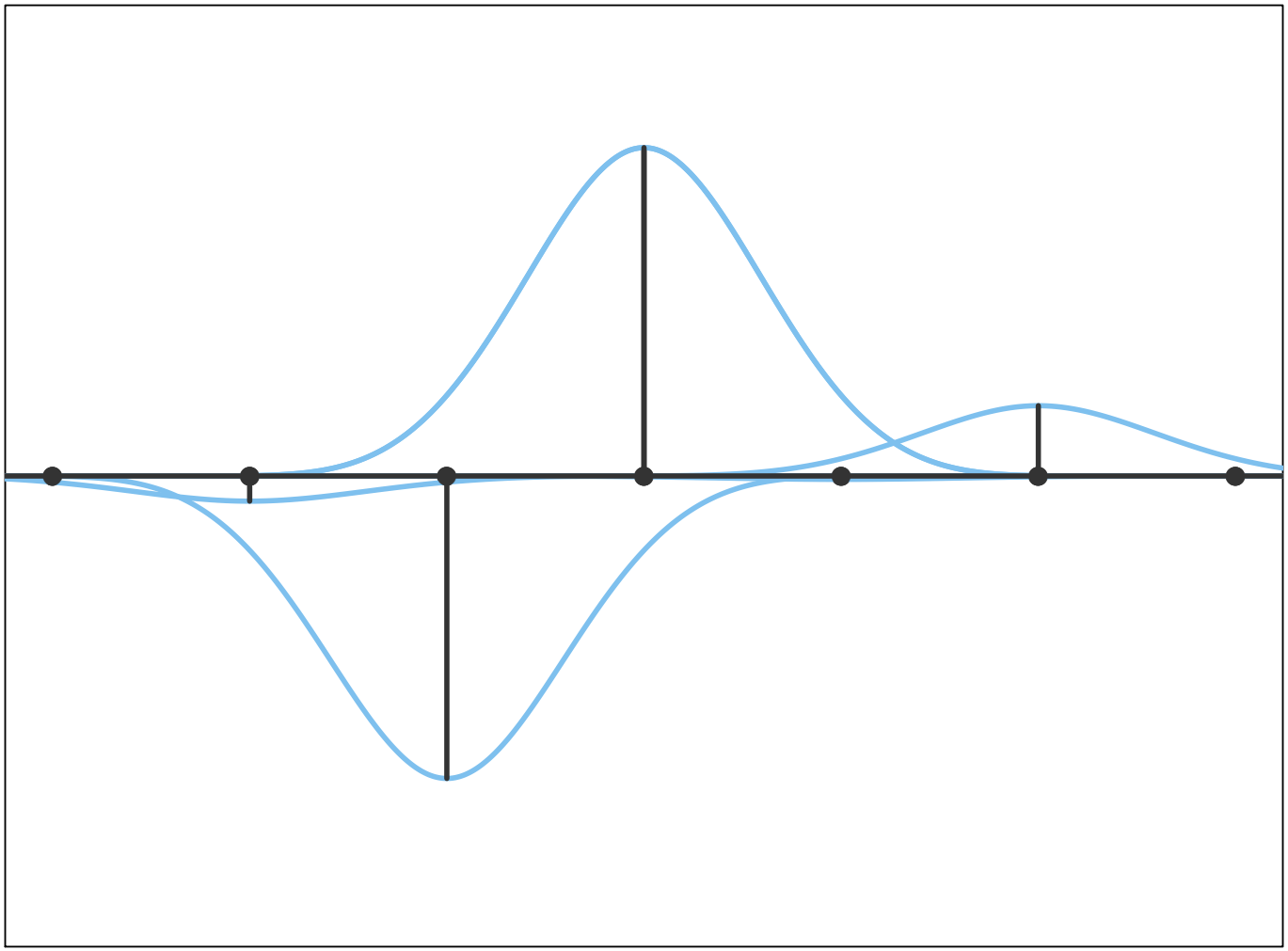
Approximating a 1-D Gaussian process

Sample basis coefficients $\mathbf{w} \sim N(\mathbf{0}, \mathbf{Q}^{-1})$ and scale each ψ_j by w_j:



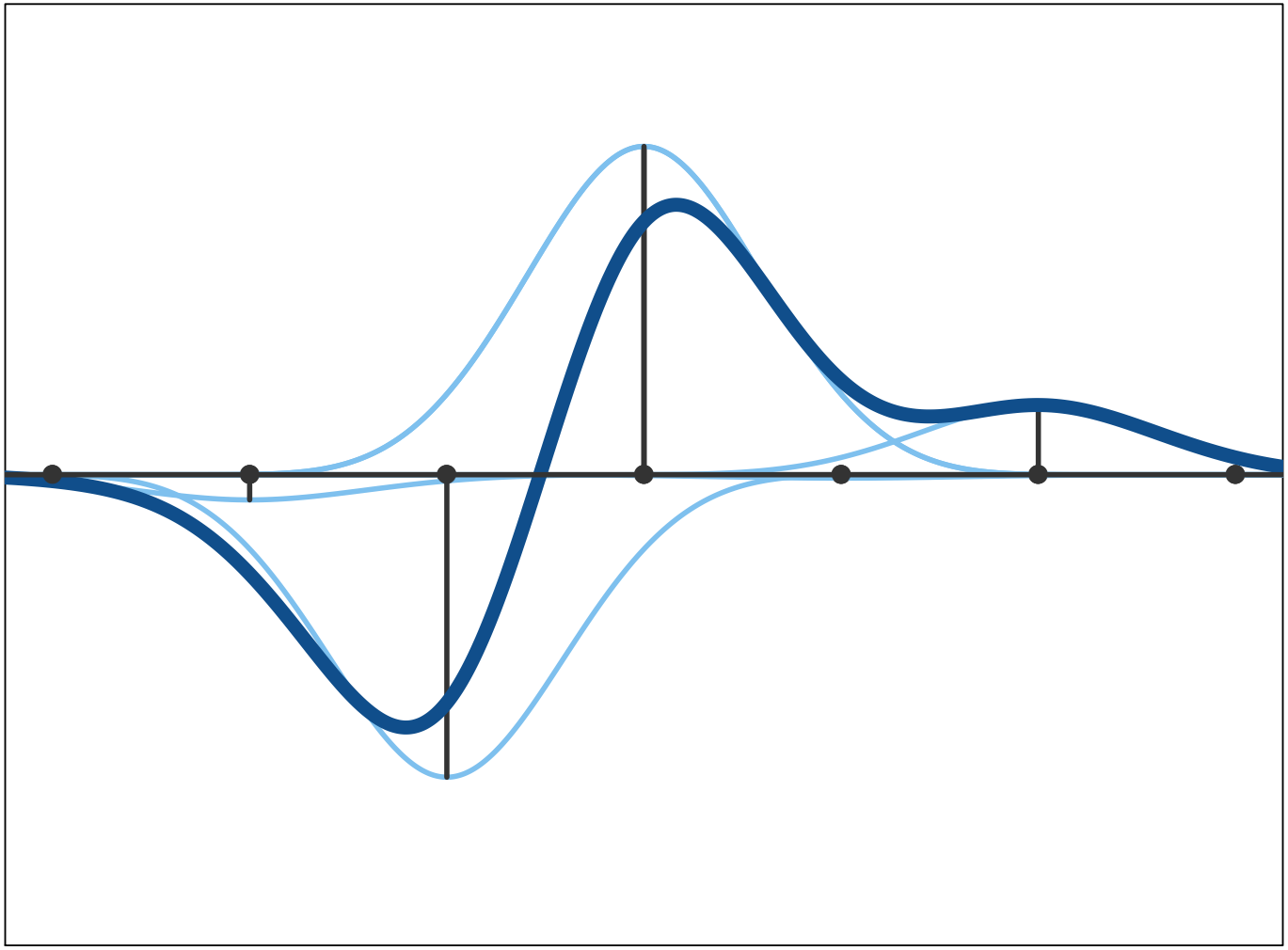
Approximating a 1-D Gaussian process

Sample basis coefficients $\mathbf{w} \sim N(\mathbf{0}, \mathbf{Q}^{-1})$ and scale each ψ_j by w_j:



Approximating a 1-D Gaussian process

Model $y(x)$ as $y(x) = \sum_{j=1}^{m} w_j \psi_j(x)$:



Spatial random effects model

Suppose

$$y(x) = \sum_{j=1}^{m} w_j \psi_j(x) \qquad \text{(basis function expansion)}$$

with

basis functions	ψ_j
basis coefficients	$\mathbf{w} \sim N(0, \mathbf{Q}^{-1})$
("spatial random effects")	

and covariance function

$$\mathrm{cov}(y(x), y(x')) = \sum_{j,k}^{m} \mathbf{Q}_{jk}^{-1} \psi_j(x) \psi_k(x')$$

Spatial random effects model

Thus, for a vector of locations $\mathbf{x} = (x_1, \ldots, x_n)$, we have

$$\mathbf{y} = (y(x_1), \ldots, y(x_n))^T = \boldsymbol{\Psi}\mathbf{w}$$

with

basis matrix $(n \times m)$	$\boldsymbol{\Psi}$ where $\boldsymbol{\Psi}_{i,j} = \psi_j(x_i)$
basis coefficients	$\mathbf{w} \sim N_m(0, \mathbf{Q}^{-1})$

Thus, $\mathbf{y}$ is normally distributed with mean zero and covariance $\boldsymbol{\Psi}\mathbf{Q}^{-1}\boldsymbol{\Psi}^T$.

Spatial random effects model

To sum up, we've constructed a Gaussian process as the sum of randomly weighted basis functions.

It turns out that if we choose the distribution of weights and the basis functions wisely, we can approximate a Matérn process while ensuring computational efficiency.

SPDE approach

Lindgren, Rue, and Lindström (2011) demonstrate that a Matérn process is a solution to a particular stochastic PDE.

Essentially, they show that by specifying a particular choice of basis functions and precision matrix $\mathbf{Q}$ for the basis function weights, they can obtain an approximate solution for this SPDE.

Moreover, this strategy produces a sparse precision matrix $\mathbf{Q}$, allowing for cheaper computation. In fact, we are modeling the basis weights $\mathbf{w}$ as a Gaussian Markov Random Field.

Recap

Loosely speaking, we would like to model temperature as following a Matérn process, but this quickly becomes computationally intractable as the number of locations increases.

Instead, we specify a set of basis functions with random weights that follow a GMRF with sparse precision matrix. In particular, we choose this GMRF so that the induced covariance function on our continuous spatial domain approximates a Matérn covariance.

Utilizing the GMRF allows us to ensure our model is computationally tractable while still allowing us the flexibility of the Matérn family.

SPDE in 1D

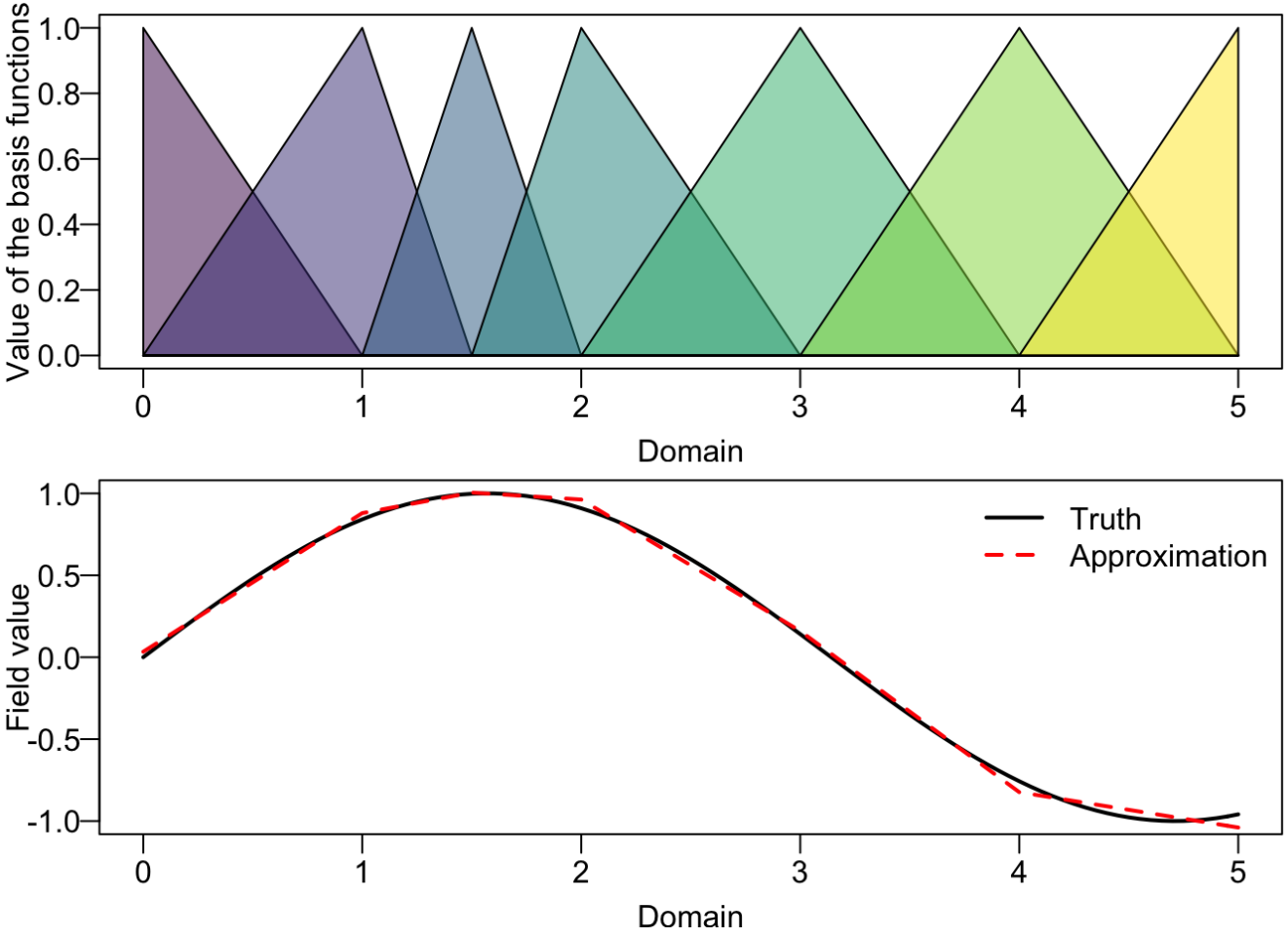


Figure 2.5 from *ASMwSPDE...*

SPDE in 2D

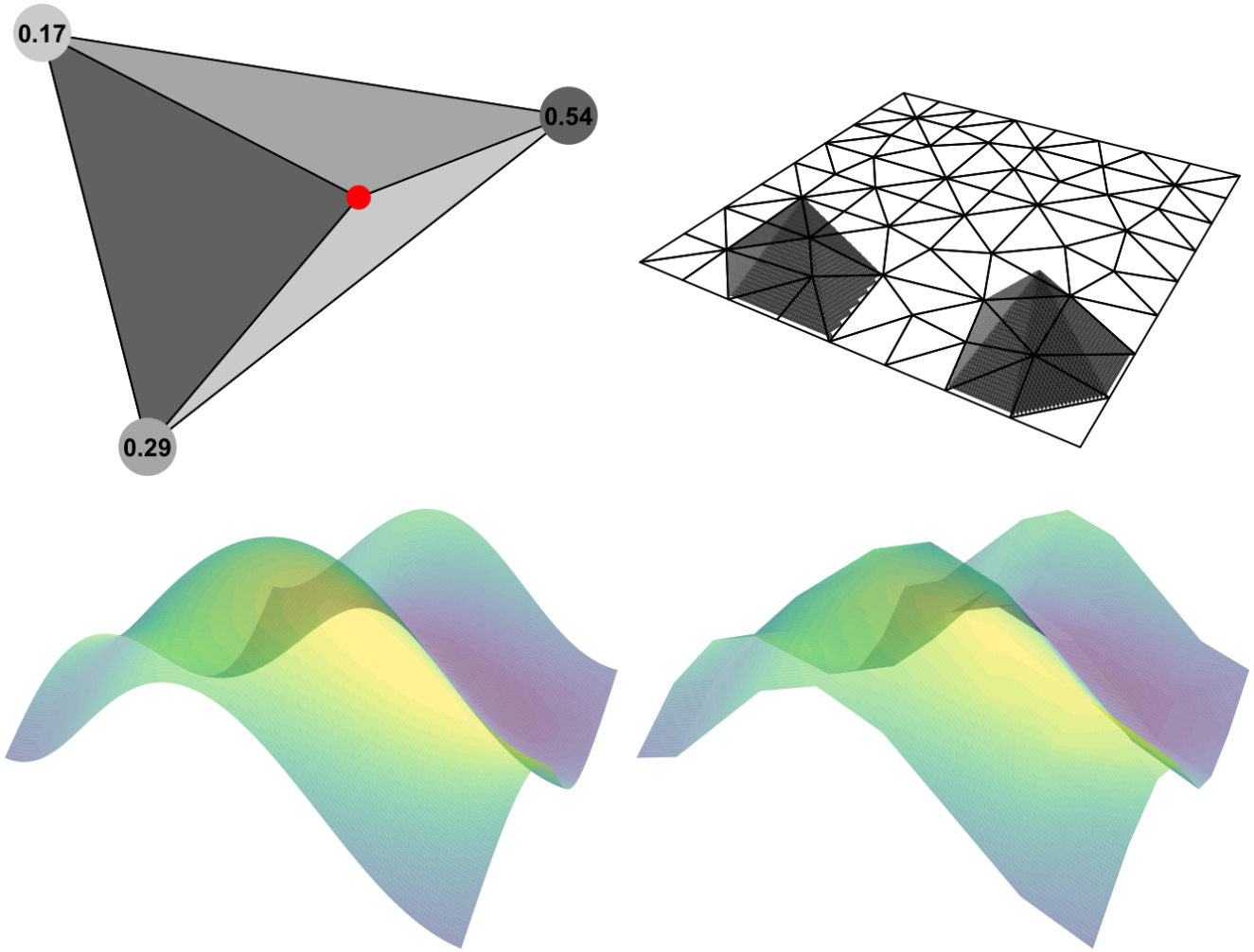


Figure 2.6 from *ASMwSPDE...*

INLA: Integrated Nested Laplace Approximation

After we specify a mesh/set of basis functions, it remains to "fit" the model to our observations.

One option is to adopt a Bayesian approach, specifying prior distributions for our fixed effect and covariance parameters. By applying Bayes' rule with our observations, we can obtain posterior distributions for our parameters and any integrated quantities.

INLA is an alternative to Markov Chain Monte Carlo (MCMC) that allows us to approximate these posterior distributions.

INLA: Integrated Nested Laplace Approximation

...come back next week to learn more!

Toy example

...time permitting

Helpful Resources

Continuous Spatial Processes:

- *Handbook of Spatial Statistics*, edited by Gelfand, Diggle, Fuentes, and Guttorp
- *Interpolation of Spatial Data* by Stein
- *Statistics for Spatial Data* by Cressie
- *Statistics for Spatio-Temporal Data* by Cressie and Wikle

Helpful Resources

SPDEs for spatial modeling:

- Lindgren, F., H. Rue, and J. Lindström. 2011. An Explicit Link Between Gaussian Fields and Gaussian Markov Random Fields: The Stochastic Partial Differential Equation Approach (with Discussion). J. R. Statist. Soc. B 73 (4): 42398.
- Bakka, H. (2018). How to solve the stochastic partial differential equation that gives a Matrn random field using the finite element method. Retrieved from https://arxiv.org/abs/1803.03765
- Simpson, D., Lindgren, F., Rue, H., 2012. Think continuous: Markovian Gaussian models in spatial statistics. Spat. Stat. 1, 1629.
- Besag, J. (1981). On a System of Two-Dimensional Recurrence Equations. Journal of the Royal Statistical Society. Series B (Methodological), 43(3), 302-309.